Student Skill-to-Job Matcher

B.Tech CSE Mini Project Documentation

Project Overview

Student Skill-to-Job Matcher is an AI-powered web application designed to help students identify suitable job roles based on their skills, interests, education, technical knowledge, and career goals. The project was developed using Base44 AI.

Problem Statement

Many college students have different technical skills but are unsure about which job roles are suitable for them. They may also lack awareness about the skills required for specific career opportunities. Students often spend a lot of time searching for suitable job roles and understanding what additional skills they need to develop. Therefore, there is a need for a simple and intelligent system that can analyze a student's profile and recommend suitable career and job roles based on their existing skills and interests.

Proposed Solution

The Student Skill-to-Job Matcher provides personalized job-role recommendations based on information provided by the student. The system considers technical skills, programming languages, education, areas of interest, previous projects, certifications, career goals, and preferred domains. Based on this information, the application provides suitable job roles and helps students understand the skills they need to improve.

Objectives

• Help students identify suitable job roles.

• Match student skills with relevant career opportunities.

• Identify skill gaps for selected job roles.

• Provide personalized career recommendations.

• Help students understand which technologies they should learn.

• Provide a simple and user-friendly career guidance platform.

Key Features

Student Profile

Students can provide name, education, branch, year of study, programming skills, technical skills, interests, projects, certifications, and career goals.

Skill-Based Recommendations

The application analyzes the student's skills and recommends suitable job roles such as Web Developer, Software Developer, Python Developer, Java Developer, Data Analyst, QA/Testing Engineer, and Database Developer.

Match Score

Recommended job roles can be displayed with a matching percentage, helping students understand how closely their profile matches a particular role.

Skill Gap Analysis

The system identifies skills students may need to improve for their selected career role.

Career Recommendations

Students can explore recommended career paths based on their interests and technical background.

Responsive Design

The application is designed for desktop, laptop, tablet, and mobile devices.

Technology Used

Technology	Purpose
Base44 AI	Application development
AI-based Recommendation Logic	Student-to-job matching
Web Technologies	User interface
Database	Storing student and recommendation information
Responsive UI	Multi-device accessibility

Working Process

→ Student Registration

→ Create Student Profile

→ Enter Skills & Interests

→ Analyze Student Profile

→ Match Skills With Job Roles

→ Generate Recommendations

→ Calculate Match Score

→ Identify Skill Gaps

→ Display Suitable Career Options

Example

Student Profile

Branch: Computer Science and Engineering
Skills: C, Java, Python, HTML, CSS
Interest: Web Development
Career Goal: Web Developer

Recommendation

Recommended Role: Web Developer
Match: 90%
Skills to Improve: JavaScript, React, Git, REST APIs

Advantages

• Easy to use

• Personalized recommendations

• Helps students choose suitable career paths

• Identifies skill gaps

• Saves time in career research

• Useful for college students and freshers

• Can be extended with more job roles and technologies

Future Enhancements

• Real-time job listings

• Internship recommendations

• Resume analysis

• AI-powered resume improvement

• Interview question recommendations

• Job application tracking

• LinkedIn profile analysis

• Personalized learning roadmaps

• Job market trend analysis

• Integration with job portals

Project Details

Project Type	Mini Project – B.Tech Computer Science and Engineering
Project Category	Artificial Intelligence / Web Application / Career Recommendation System
Development Platform	Base44 AI
Live Project	https://intel-project-path.base44.app

Conclusion

The Student Skill-to-Job Matcher is designed to bridge the gap between students' existing skills and suitable career opportunities. By analyzing student profiles and providing personalized job-role recommendations and skill-gap information, the application can help students make better career decisions and prepare themselves for the job market. This project demonstrates how AI-powered application development can be used to solve real-world problems faced by college students.

Developed By: B.Tech Computer Science and Engineering Student

Project: Student Skill-to-Job Matcher **Platform:** Base44 AI